

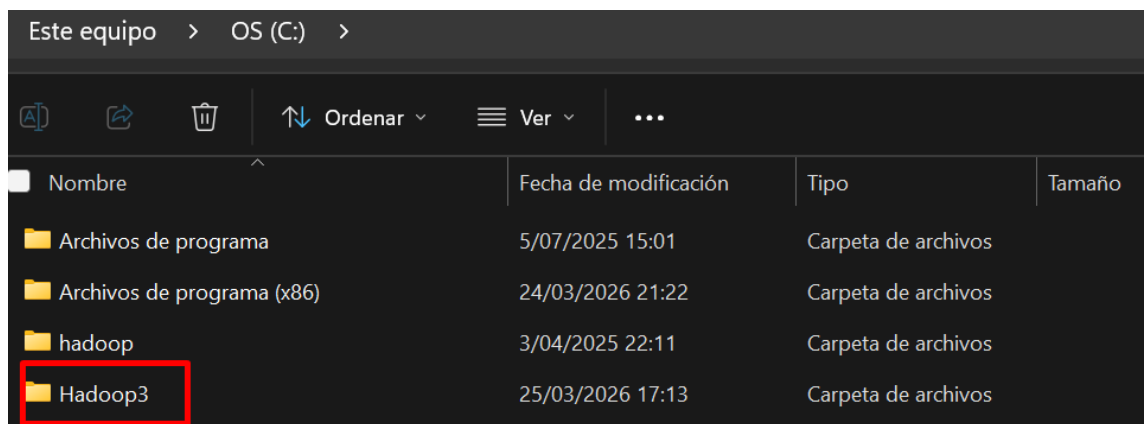
Instalación de Hadoop 3.3.0 en Windows

Prerequisitos:

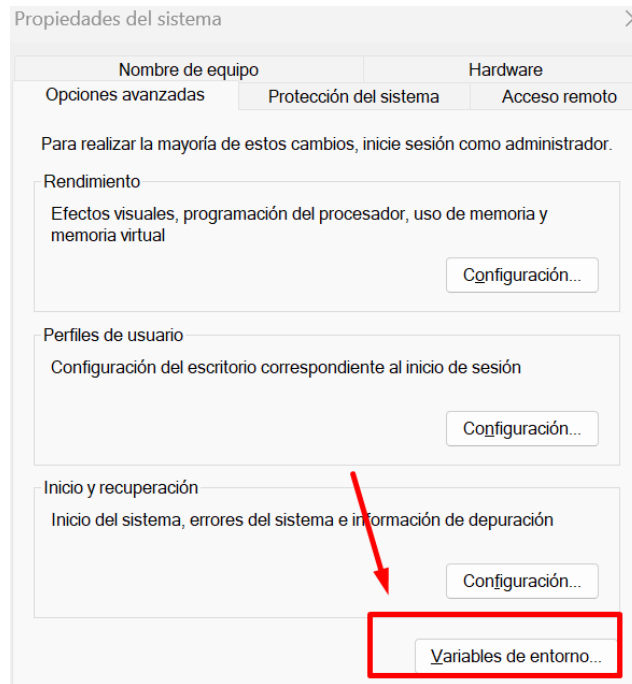
1. Java 8 runtime environment (JRE)
2. Apache Hadoop 3.3.0

Instrucciones

1. Según el Manual “Instalación de JDK” con el instalador:
https://javadl.oracle.com/webapps/download/AutoDL?BundleId=247948_0ae14417abb444ebb02b9815e2103550
Al final su instalación de Java debe estar en la ruta C:/Java
2. Descargar Hadoop 3.3.0 desde la dirección:
<https://archive.apache.org/dist/hadoop/common/hadoop-3.3.0/hadoop-3.3.0.tar.gz>
3. Extraer el contenido de hadoop 3.3.0.targz y colocarlo en el directorio C:



4. Configurar las variables de entorno de Windows:



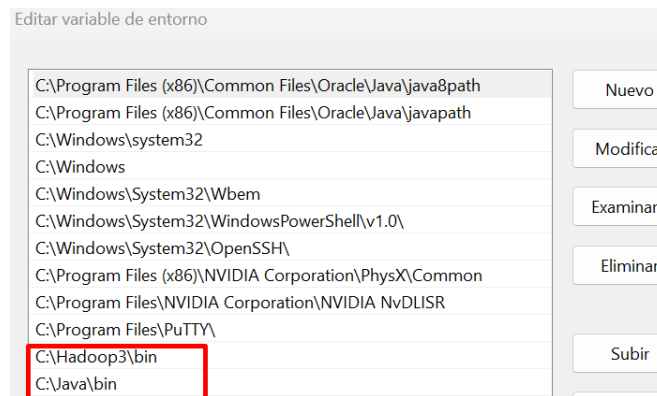
- Variable: JAVA_HOME - valor: C:\Java

This is a screenshot of the 'Edit Environment Variable' dialog box. The 'Nombre de la' (Name) field is set to 'JAVA_HOME'. The 'Valor de la' (Value) field is set to 'C:\Java'. At the bottom, there are three buttons: 'Examinar directorio...' (Browse directory...), 'Examinar archivo...' (Browse file...), and 'Aceptar' (Accept).

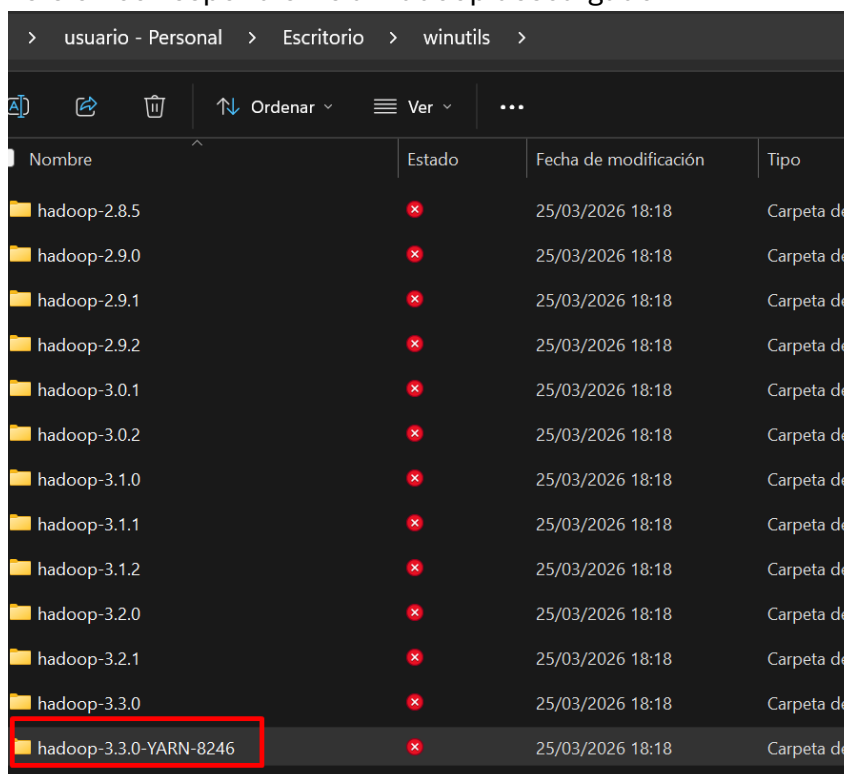
- Variable: HADOOP_HOME - valor: C:\Hadoop3 (o el nombre de tu carpeta)

This is a screenshot of the 'Edit Environment Variable' dialog box. The 'Nombre de la' (Name) field is set to 'HADOOP_HOME'. The 'Valor de la' (Value) field is set to 'C:\Hadoop3'. At the bottom, there are three buttons: 'Examinar directorio...' (Browse directory...), 'Examinar archivo...' (Browse file...), and 'Aceptar' (Accept).

- En el path agregar la ruta del bin de hadoop y de Java como muestra en la imagen



5. Descargar el archivo Winutils del aula virtual y copiar la carpeta bin de la versión correspondiente al hadoop descargado.



6. Configurar los archivos base de hadoop ubicados en %HADOOP_HOME%\etc\hadoop
 - Core-site.xml, colocar el siguiente código

```
<configuration>
  <property>
    <name>fs.default.name</name>
    <value>hdfs://0.0.0.0:19000</value>
  </property>
</configuration>
```

- En el directorio de %HADOOP_HOME% crear un directorio *data* y luego los directorios *datanode* y *namenode* y luego editar el archivo *hdfs-site.xml*

```
<configuration>
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>
  <property>
    <name>dfs.namenode.name.dir</name>
    <value>/Hadoop3/data/namenode</value>
  </property>
  <property>
    <name>dfs.datanode.data.dir</name>
    <value>/Hadoop3/data/datanode</value>
  </property>
</configuration>
```

- *Mapred-site.xml*, modificar con:

```
<configuration>
  <property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
  </property>
  <property>
    <name>mapreduce.application.classpath</name>
    <value>%HADOOP_HOME%/share/hadoop/mapreduce/*;%HADOOP_HOME%/share/hadoop/mapreduce/lib/*;%HADOOP_HOME%/share/hadoop/common/*;%HADOOP_HOME%/share/hadoop/common/lib/*;%HADOOP_HOME%/share/hadoop/yarn/*;%HADOOP_HOME%/share/hadoop/yarn/lib/*;%HADOOP_HOME%/share/hadoop/hdfs/*;%HADOOP_HOME%/share/hadoop/hdfs/lib/*</value>
  </property>
</configuration>
```

- *Yarn-site*: modificar con:

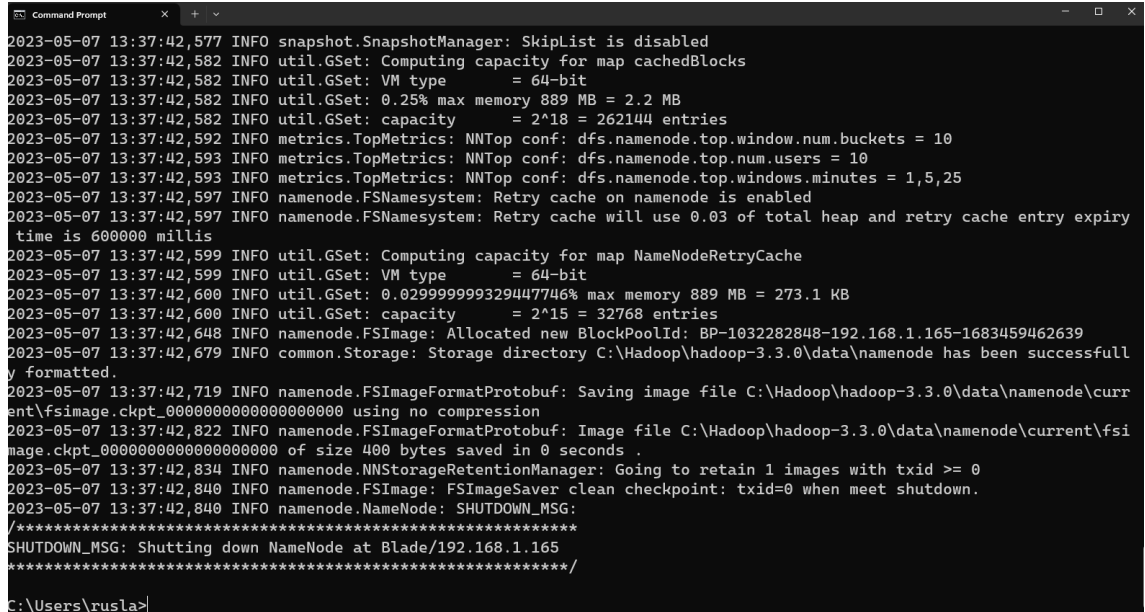
```
<configuration>
  <property>
    <name>yarn.nodemanager.aux-services</name>
    <value>mapreduce_shuffle</value>
  </property>
  <property>
    <name>yarn.nodemanager.env-whitelist</name>
    <value>JAVA_HOME,HADOOP_COMMON_HOME,HADOOP_HDFS_HOME,HADOOP_CONF_DIR,CLASSPATH_PREPEND</value>
  </property>
</configuration>
```

```
D_DISTCACHE,HADOOP_YARN_HOME,HADOOP_MAPRED
_HOME</value>
</property>
</configuration>
```

7. Aplicar el formateo de los directorios del HDFS ejecutar:

```
hdfs namenode -format
```

Si todo salió correctamente debería mostrar una salida parecida a:



```
Command Prompt
2023-05-07 13:37:42,577 INFO snapshot.SnapshotManager: SkipList is disabled
2023-05-07 13:37:42,582 INFO util.GSet: Computing capacity for map cachedBlocks
2023-05-07 13:37:42,582 INFO util.GSet: VM type = 64-bit
2023-05-07 13:37:42,582 INFO util.GSet: 0.25% max memory 889 MB = 2.2 MB
2023-05-07 13:37:42,582 INFO util.GSet: capacity = 2^18 = 262144 entries
2023-05-07 13:37:42,592 INFO metrics.TopMetrics: NNTop conf: dfs.namenode.top.window.num.buckets = 10
2023-05-07 13:37:42,593 INFO metrics.TopMetrics: NNTop conf: dfs.namenode.top.num.users = 10
2023-05-07 13:37:42,593 INFO metrics.TopMetrics: NNTop conf: dfs.namenode.top.windows.minutes = 1,5,25
2023-05-07 13:37:42,597 INFO namenode.FSNamesystem: Retry cache on namenode is enabled
2023-05-07 13:37:42,597 INFO namenode.FSNamesystem: Retry cache will use 0.03 of total heap and retry cache entry expiry
time is 600000 millis
2023-05-07 13:37:42,599 INFO util.GSet: Computing capacity for map NameNodeRetryCache
2023-05-07 13:37:42,599 INFO util.GSet: VM type = 64-bit
2023-05-07 13:37:42,600 INFO util.GSet: 0.029999999329447746% max memory 889 MB = 273.1 KB
2023-05-07 13:37:42,600 INFO util.GSet: capacity = 2^15 = 32768 entries
2023-05-07 13:37:42,648 INFO namenode.FSImage: Allocated new BlockPoolId: BP-1032282848-192.168.1.165-1683459462639
2023-05-07 13:37:42,679 INFO common.Storage: Storage directory C:\Hadoop\hadoop-3.3.0\data\namenode has been successfull
y formatted.
2023-05-07 13:37:42,719 INFO namenode.FSImageFormatProtobuf: Saving image file C:\Hadoop\hadoop-3.3.0\data\namenode\curr
ent\fsimage.ckpt_00000000000000000000 using no compression
2023-05-07 13:37:42,822 INFO namenode.FSImageFormatProtobuf: Image file C:\Hadoop\hadoop-3.3.0\data\namenode\current\fsi
mage.ckpt_00000000000000000000 of size 400 bytes saved in 0 seconds .
2023-05-07 13:37:42,834 INFO namenode.MNStorageRetentionManager: Going to retain 1 images with txid >= 0
2023-05-07 13:37:42,840 INFO namenode.FSImage: FSImageSaver clean checkpoint: txid=0 when meet shutdown.
2023-05-07 13:37:42,840 INFO namenode.NameNode: SHUTDOWN_MSG:
/*****
SHUTDOWN_MSG: Shutting down NameNode at Blade/192.168.1.165
*****/
C:\Users\rusla>
```

8. Ir a la carpeta sbin de la instalación de hadoop y ejecutar:

```
start-all.cmd
```

Revisar que se haya activado los servicios de:

- Hadoop Namenode
- Hadoop datanode
- YARN Resourc Manager
- YARN Node Manager

```

Apache Hadoop Distribution hadoop_namenode
17/07/2017 15:50:09 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:10 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:11 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:12 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:13 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:14 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:15 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:16 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:17 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:18 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:19 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:20 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:21 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:22 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:23 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:24 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:25 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:26 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:27 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:28 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:29 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:30 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:31 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:32 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:33 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:34 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:35 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:36 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:37 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:38 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:39 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:40 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:41 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:42 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:43 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:44 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:45 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:46 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:47 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:48 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:49 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:50 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:51 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:52 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:53 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:54 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:55 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:56 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:57 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:58 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:50:59 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:00 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:01 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:02 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:03 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:04 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:05 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:06 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:07 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:08 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:09 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:10 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:11 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:12 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:13 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:14 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:15 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:16 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:17 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:18 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:19 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:20 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:21 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:22 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:23 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:24 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:25 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:26 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:27 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:28 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:29 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:30 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:31 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:32 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/2017 15:51:33 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7

```

9. Abir en navegador: <http://localhost:8088/>

The screenshot displays the Hadoop Distributed File System (HDFS) web interface. The top navigation bar includes links for 'Cluster', 'About', 'Nodes', 'Node Labels', 'Applications', 'NEW', 'NEW SAVING', 'SUBMITTED', 'ACCEPTED', 'RUNNING', 'FINISHED', 'FAILED', 'KILLED', and 'Scheduler'. The main content area is titled 'All Applications' and shows a table of cluster metrics. The metrics table has columns for 'Apps Submitted', 'Apps Pending', 'Apps Running', 'Apps Completed', 'Containers Running', 'Memory Used', 'Memory Total', 'Memory Reserved', 'VCores Used', 'VCores Total', and 'VCores Reserved'. The values for all these metrics are 0. Below the metrics table, there is a section for 'Cluster Nodes Metrics' with columns for 'Active Nodes', 'Decommissioning Nodes', 'Decommissioned Nodes', 'Lost Nodes', 'Unhealthy Nodes', 'Rebooted Nodes', and 'Shutdown Nodes'. All these values are also 0. The 'Scheduler Metrics' section shows 'Scheduler Type' as 'Capacity Scheduler', 'Scheduling Resource Type' as '[MEMORY]', 'Minimum Allocation' as '<memory:1024, vCores:1>', 'Maximum Allocation' as '<memory:8192, vCores:4>', and 'Maximum Cluster Application Priority' as 0. The 'Applications' table is empty, showing 0 entries.

Referencias:

<https://github.com/ruslanmv/How-to-install-Hadoop-on-Windows/tree/master?tab=readme-ov-file>

Manual de instalación de Hadoop 2.8.0 en el aula virtual.

En Linux: <https://es.unixlinux.online/qu/1002014012.html>